

Matlab L-ven

Homework 3 - Intro to DSP

1. Even & Odd Functions: sequence

$x[n]$ is said to be even

if $x[-n] = x[n]$, and odd if

$x[-n] = -x[n]$, $n = 0, 1, 2, \dots, N-1$

$\rightarrow$ indexing is carried out modulo N

a) $[0, 1, 2, 1] \rightarrow N=4$

$$x[0] = 0, x[1] = 1, x[2] = 2, x[3] = 1$$

even:

$$n=0: x_{\text{even}}[0] = \frac{x[0] + x[-0]}{2} = \frac{0+0}{2} = \boxed{0}$$

$$n=1: x_{\text{even}}[1] = \frac{x[1] + x[-1]}{2} = \frac{2}{2} = \boxed{1}$$

$$n=2: \boxed{2}$$

$$n=3: \boxed{1}$$

$$x_{\text{even}} = [0, 1, 2, 1]$$

$$x_{\text{odd}} = x - x_{\text{even}} = [0-0, 1-1, 2-2, 1-1] = [0, 0, 0, 0]$$

→ the sequence is purely even

$$b) x[0, 1, 0, -1] \quad N=4$$

$$x[0] = 0, x[1] = 1, x[2] = 0, x[3] = -1$$

x_{even}

$$n=0: x_{\text{even}}[0] = \frac{0+0}{2} = 0$$

$$n=1: x_{\text{even}}[1] = \frac{x[1] + x[3]}{2} = \frac{1+(-1)}{2} = 0$$

$$n=2: \frac{x[2] + x[2]}{2} = 0$$

$$n=3: \frac{x[3] + x[1]}{2} = \frac{-1+1}{2} = 0$$

$$X_{\text{even}} = [0, 0, 0, 0]$$

$$* X_{\text{odd}} = X - X_{\text{even}} = [0, 1, 0, -1]$$

→ Sequence is purely odd

$$c) X = [0, 1, 2, 3] \quad N = 4$$

$$X[0] = 0, X[1] = 1, X[2] = 2, X[3] = 3$$

$X_{\text{even}} :$

$$n = 0 : X_{\text{even}}[0] = \frac{0+0}{2} = 0$$

$$n = 1 : \frac{X[1] + X[3]}{2} = 2$$

$$n = 2 : 2$$

$$n = 3 : 2$$

$$X_{\text{even}} = [0, 2, 2, 2]$$

$$X_{\text{odd}} = X - X_{\text{even}} = [0, 1-2, 2-2, 3-2] = [0, -1, 0, 1]$$

$$* \in \{ \in \in \in \}$$

$$X_{\text{odd}}[3] = 1 = -X_{\text{odd}}[1], \text{ and}$$

$$X_{\text{odd}}[2] = 0 \quad \text{is self consistent}$$

$$X_{\text{even}} = [0, 2, 2, 2],$$

$$X_{\text{odd}} = [0, -1, 0, 1]$$

$$d) \quad x(n) = n^2, \quad n = 0, 1, \dots, N-1$$

$$x[-n] = x[N-n] = (N-n)^2 \quad \text{for}$$

$$1 \leq n \leq N-1, \text{ and } x[-0] = x[0] = 0$$

$$\text{even part } x_e[n] = \frac{x[n] + x[-n]}{2} \quad ;$$

$$n=0 : x_e[0] = \frac{0+0}{2} = 0$$

$$1 \leq n \leq N-1 :$$

$$x_e[n] = \frac{n^2 + (N-n)^2}{2} = \frac{n^2 + (N^2 - 2Nn + n^2)}{2} = n^2 - Nn + \frac{N^2}{2}$$

$$\text{odd part } x_o[n] = \frac{x[n] - x[-n]}{2} \quad ;$$

$$n=0 : x_o[0] = \frac{0-0}{2} = 0$$

$$1 \leq n \leq N-1 :$$

$$x_o[n] = \frac{n^2 - (N-n)^2}{2} = \frac{n^2 - (N^2 - 2Nn + n^2)}{2} = Nn - \frac{N^2}{2}$$

$$x_{\text{even}}[0] = 0, \quad x_{\text{even}}[n] = n^2 - Nn + \frac{N^2}{2} \quad \text{for } 1 \leq n \leq N-1$$

$$x_{\text{odd}}[0] = 0, \quad x_{\text{odd}}[n] = Nn - \frac{N^2}{2} \quad \text{for } 1 \leq n \leq N-1$$

2) DFT Symmetry if $Y[k]$ denotes the k th element of the length N DFT of y

a) y real $\Leftrightarrow Y[k] = Y[N-k]^*$ (Hermitian)

($\Rightarrow$) if $y[n] \in \mathbb{R}$:

$$Y[N-k] = \sum_n y[n] e^{-j \frac{2\pi}{N} (N-k)n}$$

$$\sum_n y[n] e^{+j \frac{2\pi}{N} kn} = \left(\sum_n y[n] e^{-j \frac{2\pi}{N} kn} \right)^* = Y[k]^*$$

So $Y[N-k] = Y[k]^*$ (equivalently $Y[k] =$

$$Y[N-k]^*$$

($\Leftarrow$) if $Y[N-k] = Y[k]^*$ for all k ,
then

$$y[n]^* = \frac{1}{N} \sum_k Y[k]^* e^{-j \frac{2\pi}{N} kn} =$$

$$\frac{1}{N} \sum_k Y[N-k] e^{-j \frac{2\pi}{N} kn}$$

$$= \frac{1}{N} \sum_m Y[m] e^{+j \frac{2\pi}{N} mn} = y[n],$$

so $y[n]$ is real

$$b) \text{ } y \text{ imaginary} \Leftrightarrow Y[k] = -Y[N-k]^*$$

(anti-Hermitian)

$$\Rightarrow Y[N-k] = \sum_n y[n] e^{+j \frac{2\pi}{N} kn} =$$

$$\left(\sum_n y[n]^* e^{-j \frac{2\pi}{N} kn} \right)^* =$$

$$\sum_n (-y[n]) e^{-j \frac{2\pi}{N} kn} =$$

$$(-Y[k])^* = -Y[k]^*$$

so $Y[N-k] = -Y[k]^*$

($\Leftarrow$) if $Y[N-k] = -Y[k]^*$, then

$$y[n]^* = \frac{1}{N} \sum_k Y[k]^* e^{-j\frac{2\pi}{N}kn} =$$

$$\frac{1}{N} \sum_k (-Y[N-k]) e^{-j\frac{2\pi}{N}kn} =$$

$$-y[n],$$

hence y is purely imaginary

$$(y[n]^* = -y[n])$$

$$c) \quad y \text{ even} \Leftrightarrow Y \text{ even}$$

$$\text{Even means } y[-n] = y[n]$$

($\Rightarrow$)

$$Y[N-k] = \sum_n y[n] e^{-j \frac{2\pi}{N} (N-k)n} =$$

$$\sum_n y[n] e^{+j \frac{2\pi}{N} kn} = \sum_n y[-n] e^{-j \frac{2\pi}{N} kn}$$

$$\stackrel{n=-n}{=} \sum_m y[m] e^{-j \frac{2\pi}{N} km} = Y[k]$$

So $Y[N-k] = Y[k]$ i.e. Y is even

$\Leftarrow$ if $Y[N-k] = Y[k]$ for all k ,

apply same logic to the inverse DFT

to get $y[-n] = y[n]$

$$d) \quad y \text{ odd} \Leftrightarrow Y \text{ odd}$$

$$\text{odd means } y[-n] = -y[n]$$

$\Rightarrow$

$$Y[N-k] = \sum_n y[-n] e^{-j \frac{2\pi}{N} kn} = - \sum_n y[n] e^{-j \frac{2\pi}{N} kn} = -Y[k]$$

$$\Leftrightarrow \text{if } Y[N-k] = -Y[k],$$

$$y[-n] = \frac{1}{N} \sum_k Y[k] e^{-j \frac{2\pi}{N} kn}$$

$$= \frac{1}{N} \sum_n Y[N-m] e^{+j \frac{2\pi}{N} mn}$$

$$\frac{1}{N} \sum_m (-Y[m]) e^{+j \frac{2\pi}{N} mn} = -y[n]$$

$$e) \quad y \text{ real, even} \Leftrightarrow Y \text{ real, even}$$

$$\text{from (a): real } y \Rightarrow Y[N-k] = Y[k]^*$$

$$\text{from (c): even } y \Rightarrow Y[N-k] = Y[k]$$

$$\text{Conclude } Y[k] = Y[k]^* \text{ (real) and}$$

$$Y[N-k] = Y[k] \text{ (even)}$$

$$f) \quad y \text{ real, odd} \Leftrightarrow Y \text{ imaginary, odd}$$

$$\text{real } y \Rightarrow Y[N-k] = Y[k]^* \text{ (a)}$$

$$\text{odd } y \Rightarrow Y[N-k] = -Y[k] \text{ (d)}$$

$$\text{Hence } Y[k]^* = -Y[k], \text{ so } Y[k]$$

is purely imaginary - and from (d) it is odd.

$$g) \quad y \text{ imaginary, even} \Leftrightarrow Y \text{ imaginary, even}$$

$$\text{imaginary } y \Rightarrow Y[N-k] = -Y[k]^* \text{ (b)}$$

$$\text{Even } y \Rightarrow Y[N-k] = Y[k] \text{ (c)}$$

$$\text{Thus } Y[k] = -Y[k]^* \text{ (purely imaginary)}$$

and, from (c) we have

$$h) \quad y \text{ imaginary, odd} \Leftrightarrow y \text{ real, odd}$$

$$\text{imaginary } y \Rightarrow Y[N-k] = -Y[k]^* \quad (b)$$

$$\text{odd } y \Rightarrow Y[N-k] = -Y[k] \quad (d)$$

$$\text{Hence } -Y[k]^* = -Y[k] \Rightarrow Y[k] = Y[k]^*$$

(real), from (d) it is odd

3) DFT: shift, flip, Linearity ($N=8$)

let

$$X[k] = \sum_{n=0}^7 x[n] e^{-j \frac{2\pi}{8} kn}, \quad k=0, \dots, 7$$

a) $x_1 = [1, 0, 0, 0, 0, 0, 0, 0]$

$$X_1[k] = \sum_{n=0}^7 x_1[n] e^{-j \frac{2\pi}{8} kn} = 1 \cdot e^0 = 1, \quad k=0, \dots, 7.$$

$$X_1[k] = 1 \text{ for all } k$$

b) $x_2 = [0, 1, 0, 0, 0, 0, 0, 0]$ use shift and express x_2 in x_1 .

time shift: $x_2[n] = x_1[(n-1) \bmod 8] \Rightarrow X_2[k] = e^{-j \frac{2\pi}{8} k} X_1[k]$

since $X_1[k] = 1$

$$X_2[k] = e^{-j \frac{2\pi}{8} k} = e^{-j \frac{\pi}{4} k}, \quad k=0, \dots, 7.$$

c) $x_3 [0, 0, 0, 0, 0, 0, 0, 1]$ use FLIP and express x_3 in x_2

Circular flip: $x_3 [n] = x_2 [(-n) \bmod 8] \Rightarrow$

$$x_3 [k] = x_2 [(-k) \bmod 8]$$

with $x_2 [k] = e^{-j\frac{\pi}{4}k}$

$$x_3 [k] = e^{+j\frac{\pi}{4}k}, \quad k = 0, \dots, 7.$$

equivalently $x_3 [k] = x_2 [8 - k]$

d) $x_4 = [2, 1, 0, 0, 0, 0, 0, 1]$ use linearity; express x_4 in x_1, x_2, x_3

decompose:

$$x_4 [n] = 2x_1 [n] + x_2 [n] + x_3 [n] \Rightarrow x_4 [k] = 2x_1 [k] + x_2 [k] + x_3 [k]$$

Substitution:

$$X_4[k] = 2 \cdot 1 + e^{-j\frac{\pi}{4}k} + e^{+j\frac{\pi}{4}k} = \\ 2 + 2 \cos\left(\frac{\pi}{4}k\right)$$

$$X_4[k] = 2 + 2\cos\left(\frac{\pi}{4}k\right), \quad k = 0, \dots, 7.$$

numerically: $k=0: 4, 1: 2+\sqrt{2}, 2: 2, 3: 2-\sqrt{2},$
 $4: 0, 5: 2-\sqrt{2}, 6: 2, 7: 2+\sqrt{2}$

e) Express $X_4(k)$ in terms of $X_1(k)$ and $X_2(k)$ only

from (c): $X_3[k] = X_2[8-k]$. from (d):

$$X_4[k] = 2X_1[k] + X_2[k] + X_2[8-k], \quad k = 0, \dots, 7$$

(with $X_1[k] = 1$, this equals $2 + e^{-j\frac{\pi}{4}k} + e^{+j\frac{\pi}{4}k}$)

2) Signal Metrics (for $y_1 = [1, 2, -3]$, $y_2 = [1, 1, 1-1]$)

definitions for a length- N sequence

$y[n]$:

$$\mu_y = \frac{1}{N} \sum_{n=0}^{N-1} y[n], \quad E_y = \sum_{n=0}^{N-1} |y[n]|^2,$$

$$P_y = \frac{1}{N} \sum_{n=0}^{N-1} |y[n]|^2, \quad \sigma_y^2 = \frac{1}{N} \sum_{n=0}^{N-1} |y[n] - \mu_y|^2$$

$$y_1 = [1, 2, -3] \quad (N=3)$$

(a) mean μ_{y_1} :

$$\mu_{y_1} = \frac{1+2+(-3)}{3} = 0$$

(b) total energy E_{y_1} :

$$|1|^2 = 1, |2|^2 = 4, |-3|^2 = 9 \Rightarrow E_{y_1} = 1 + 4 + 9 = 14$$

(c) Average power P_{y_1} :

$$P_{y_1} = \frac{1}{3} (1 + 4 + 9) = \frac{14}{3}$$

(d) Sample variance σ_{y1}^2 :

$$\sigma_{y1}^2 = \frac{1}{3} (|1-0|^2 + |2-0|^2 + |-3-0|^2) = \frac{14}{3}$$

$$y = [1, 1, 1-j] \quad (N=3)$$

(a) mean μ_{y2} :

$$\mu_{y2} = \frac{(+1) + (1-j)}{3} = \frac{2}{3}$$

(b) Total energy E_{y2} :

$$|1|^2 = 1, |1|^2 = 1, |1-j|^2 = 1^2 + (-1)^2 = 2 \Rightarrow E_{y2} = 1 + 1 + 2 = 4$$

(c) Average power P_{y2} :

$$P_{y2} = \frac{1}{3} (1 + 1 + 2) = \frac{4}{3}$$

(d) Sample variance σ_{y2}^2
deviations from $\mu_{y2} = \frac{2}{3}$

$$1 - \frac{2}{3} = \frac{1}{3} \Rightarrow \left| \frac{1}{3} \right|^2 = \frac{1}{9},$$

$$j - \frac{2}{3} = -\frac{2}{3} + j \Rightarrow \left| -\frac{2}{3} + j \right|^2 = \frac{4}{9} + 1 = \frac{13}{9},$$

$$(1-j) - \frac{2}{3} = \frac{1}{3} - j \Rightarrow \left| \frac{1}{3} - j \right|^2 = \frac{1}{9} + 1 = \frac{10}{9}$$

$$Sum = \frac{1+13+10}{9} = \frac{24}{9} = \frac{8}{3}$$

$$\sigma^2 = \frac{1}{3} \cdot \frac{10}{3} = \frac{10}{9}$$

* find the variance:

$$y_1: \mu = 0, \quad E = 14, \quad P = \frac{14}{3}, \quad \sigma^2 = \frac{14}{3}$$

$$y_2: \mu = \frac{2}{3}, \quad E = 4, \quad P = \frac{4}{3}, \quad \sigma^2 = \frac{8}{9}$$